

IMAM MEHEDI HASAN

Electrical & Electronic Engineering Student — Robotics & Automation

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PROFILE

Electrical & Electronic Engineering student at Jatiya Kabi Kazi Nazrul Islam University, focused on embedded systems, robotics, and practical automation. Builds working prototypes across mobility robots, IoT/home automation, biomedical sensing, and power electronics, with 16 independent and competition projects completed to date.

EDUCATION

B.Sc. in Electrical & Electronic Engineering (EEE) | *Jatiya Kabi Kazi Nazrul Islam University — Present*

Higher Secondary Certificate (HSC) | *Agricultural University College, Mymensingh*

Junior & Secondary School Certificate (JSC / SSC) | *Polisha High School*

Primary School Certificate (PSC) | *Protiva Pre-Cadet & Academy*

SKILLS

- Programming: Python, Embedded C/C++, Arduino
- Hardware: Microcontrollers (AVR/ESP32), PCB & Circuit Design, Sensor Integration
- Systems: Robotics & Mechatronics, Control Systems, IoT Systems, Power Electronics
- Modules: MediaPipe/Pose Detection, GPS/GSM, Biomedical Sensors (EMG/ECG)

PROJECTS

Human Following Robot

Python + MediaPipe pose-mesh tracking to follow a person autonomously.

Line Following & Obstacle Avoidance Robot

Line-tracking with real-time obstacle rerouting.

Soccer Bot

Competition robot built for robotic soccer contests.

Hexapod

Six-legged walking robot for uneven terrain.

Wall Climbing Robot

Suction-based vertical wall climbing robot.

Robotic Hand

Mechanical hand mimicking human finger movement.

Home Automation

Digitalized home appliances with remote control.

Smart Irrigation System

Automated watering based on soil/environment data.

Smart Street Light Control

Ambient-light-based automatic street lighting.

Smart Electricity Theft Detection

Detects grid theft and sends GPS-tagged alerts to substations.

Smart Fingerprint Attendance System

Biometric attendance logging.

Safety Car

Vehicle concept with layered passenger safety systems.

EMG-Based Alcohol Detection & Signaling

EMG sensor detects driver impairment and signals in-cabin.

Remote ECG Monitoring

Real-time remote cardiac monitoring system.

Emergency Alert System

Rapid-response emergency alert dispatch.

Piezoelectric Power Generation & Storage

Converts mechanical vibration into stored electrical energy.